

Kevin Le

Systems Engineer

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EDUCATION

University of California, Davis

Davis, CA

Master of Science in Computer Engineering | GPA: 4.0

Sep 2025 – Dec 2026

University of California, Davis

Davis, CA

Bachelor of Science in Computer Engineering | GPA: 3.75

Sep 2021 – Jun 2025

- Deans Honor's List — Tau Beta Pi Scholar

EXPERIENCE

AggieWorks: RoomU Team

Sep 2023 – Dec 2024

Software Engineering Intern

- Architected and implemented critical messaging infrastructure using React and TypeScript, optimizing state management to support real-time user interactions
- Worked within a small cross-functional team of developers and designers to reform the application's features and improve user retention
- Engineered SQL database-side page-caching protocols that reduced latency by 65%, directly improving user retention through optimized query execution

UC Davis College of Engineering

Sep 2024 – Mar 2025

Lab Teaching Assistant

- Mentored 60+ students in circuit analysis and hardware debugging; demonstrated mastery of signal integrity and oscilloscope operation to ensure lab accuracy
- Taught proper oscilloscope operation, breadboard best practices, and conducted lab demos to explain lecture materials taught

PROJECTS

ML Framework in CUDA

- Engineered a custom ML framework in CUDA C++, manually implementing forward and backward propagation kernels to replicate PyTorch architecture performance
- Profiled kernel execution to diagnose bottlenecks in Streaming Multiprocessor (SM) utilization and memory throughput, identifying and minimizing high-latency global memory accesses
- Optimized memory patterns to maximize bandwidth efficiency, achieving 98% classification accuracy and training speeds comparable to highly-optimized PyTorch libraries

GeMM Hardware Accelerator | Verilog, FPGA, ModelSim

- Architected and synthesized a high-performance General Matrix Multiplier (GeMM) engine in Verilog, optimizing for parallel data processing on an FPGA
- Conducted rigorous timing and functional verification using ModelSim and Quartus, identifying and resolving critical path delays to ensure timing closure at target clock frequencies.
- Engineered a multi-threaded compute fabric consisting of 4 independent Multiply-Accumulate (MAC) processing units, achieving a 500% throughput increase over baseline serial implementations

Distributed Computing using MPI

- Engineered a distributed C application to render the Mandelbrot Set, utilizing OpenMPI for parallel processing across a multi-node cluster
- Optimized inter-node communication by implementing non-blocking MPI primitives and collective operations, achieving a 3x throughput speedup by minimizing synchronization bottlenecks
- Automated node orchestration using custom Bash scripts to dynamically provision available compute resources and scale processing power on demand

RELEVANT SKILLS

Languages: C, C++ (C++20), Python, Verilog, RISC-V, Go

Developer Tools: Git, CMake, gdb, Kubernetes, Docker, Cacti 7, Accel-Sim, Nsight Compute (NCU), Quartus, ModelSim

Miscellaneous: CUDA, Linux, PyTorch, OpenMPI, OpenMP, OpenCL, FPGA